

01 – THE REAL QUESTION

What the Ministry is **actually** asking

Before explaining the technology, let's state the three fears the Ministry has – and address each one directly.



FEAR 01

Can someone hack this and change a property record?

Answer: No. Because property records are never stored only in one place. The system uses a method called **cryptographic hashing** — the same technology used by SBI, HDFC, and every Indian bank for internet banking. Any change to any record — even one character — creates a completely different hash, which immediately triggers an alert. You cannot silently edit a record. The tamper is always visible.

SHA-256 Hashing

Same as Indian Banking



FEAR 02

Is blockchain legal in India? Can it be used as evidence?

Answer: We don't use blockchain as a legal record. We use it as an **audit infrastructure layer** — meaning it is the system that logs who did what, when, and what it looked like at that moment. It is an internal tamper-proof logbook. The legal record remains the Sub-Registrar's office, as it always has been. We add a shadow copy that cannot be altered. Courts already accept digital evidence under **Section 65B, Indian Evidence Act**.

Audit Log Only

Section 65B Evidence Act

Not Replacing Registrar



FEAR 03

What if the BPC company itself turns corrupt? Can insiders manipulate it?

Answer: No single person — including BPC's own engineers — can change a record without leaving an irremovable trace. The system is designed so that every action by every user, admin, or officer creates a permanent log entry. Deleting a log entry is technically impossible because logs are written across multiple independent nodes simultaneously. The government's own designated node is one of those nodes. **Government is a co-owner of the audit trail.**

Multi-Node Architecture

Govt Node = Co-Custodian

The **seven walls** a hacker must break – simultaneously

BPC is not protected by one security system. It uses 7 independent layers. To succeed, an attacker must break all 7 at the same time. That is computationally and practically impossible.

LAYER 1 Identity	Aadhaar OTP Login	Who you are is verified before you touch any data. Officers have different access levels than citizens.
	Biometric Verification	
	Role-Based Access (RBAC)	
	Device Fingerprinting	
	Session Tokens (JWT)	
LAYER 2 Transmission	TLS 1.3 Encryption	All data moving between users and servers is encrypted. Same standard as NPCI (UPI). Cannot be intercepted.
	Certificate Pinning	
	API Gateway (Rate Limiting)	
	IP Whitelisting (Govt nodes)	
LAYER 3 Application	Input Validation (SQL Injection Block)	The software itself is hardened. All known attack types are blocked at code level before they reach data.
	OWASP Top 10 Compliance	
	AI Document Forgery Detection	
	Automated Pen Testing (quarterly)	
AES-256 Encryption at Rest		



KEY POINT FOR MINISTRY

These 7 layers are **independent**. Failure of one does not compromise the system. This is called "**Defense in Depth**" — the same architecture used by NPCI for UPI, by UIDAI for Aadhaar, and by Indian banks for core banking. BPC follows the exact same security standard as existing trusted government infrastructure.

03 — BLOCKCHAIN: WHAT IT DOES AND WHY

Blockchain, explained for a Minister

No technical jargon. Here is the exact logic of what blockchain does in BPC, step by step, using a property transaction as the example.

▲ WHAT BLOCKCHAIN IS NOT IN BPC

It is **not the legal title record**. It is **not a cryptocurrency**. It is **not replacing the Sub-Registrar**. It is a **permanent, distributed logbook** — like a notarized diary that is simultaneously held by multiple trusted parties, and cannot be edited after writing.

// REAL EXAMPLE: WHAT HAPPENS WHEN RAMESH SELLS HIS FLAT TO PRIYA

Property:	Flat 402, Sunrise Tower, Andheri, Mumbai
BPC-ID:	BPC-MH-2024-00847361
Sale Deed Hash:	a3f8d1c9e2b7...4f92a1 (256-bit fingerprint of the document)
Block Written:	{ Seller: Ramesh, Buyer: Priya, Date: 14-Mar-2025 14:32:07 IST, DeedHash: a3f8d1c9e2b7...4f92a1, SubRegistrar: verified ✓, Officer-Sig: [digital signature] }
Stored at:	BPC Server (Mumbai) + NIC Server (Delhi) + State Govt Node (Maharashtra)

If someone tries to
change "Priya" to "Fraud
Name":

Hash becomes: 9x2k17pq... [COMPLETELY DIFFERENT] → All 3 nodes
immediately detect mismatch → Alert triggered → Change rejected
→ Fraud logged



STEP 01 - WHY HASH?

Every document gets a unique fingerprint

A "hash" is a mathematical fingerprint of a document. Change even one full-stop in the sale deed, and the fingerprint changes completely. It is **mathematically impossible to produce the same fingerprint from two different documents**. This is called SHA-256 — the same algorithm used to secure your UPI transactions right now.

SHA-256

Same as UPI Security



STEP 02 - WHY DISTRIBUTED?

The same logbook is held by 3 independent parties

BPC uses a **permissioned private blockchain** — meaning only approved nodes can participate. These nodes are: **(1) BPC's server, (2) NIC / State Government's server, (3) An independent audit node**. All three must agree that a record is correct. If BPC's server is compromised, the government's copy exposes the fraud immediately. **No single entity can manipulate the record alone.**

Permissioned Blockchain

Govt Node = Co-Custodian



STEP 03 - WHY CHAINED?

Each block references the one before it — like a chain of wax seals

Think of old-style official letters where each page is stamped with the previous page's seal. If you remove or change one page, the chain breaks and everyone can see it. Each new blockchain entry contains a reference to the previous block's hash. **You cannot insert, delete, or modify any past entry without breaking the entire chain**, which is immediately detectable by all nodes.

Immutable Chain

Tamper-Evident by Design



STEP 04 — LEGAL STANDING

This is evidence, not title. Here's the legal basis.

The blockchain log in BPC is treated as a **tamper-evident electronic record** under **Section 2(1)(t) of the IT Act, 2000**. Electronic records are admissible as evidence in Indian courts under **Section 65B of the Indian Evidence Act** with a proper certificate. We do not need blockchain to be "legalized as title" — we use it as a **certified audit trail**, which Indian law already recognizes.

IT Act 2000 — S.2(1)(t)

Evidence Act — S.65B

ANALOGY FOR THE MINISTER

Imagine every property transaction is photographed, notarized, and three sealed copies are sent to three different trusted government offices. If anyone tampers with one copy, the other two immediately expose the fraud. **That is exactly what BPC's blockchain layer does — digitally, automatically, at zero human cost.**

04 — THE DATA CHALLENGE & STRATEGY

The **hardest problem**: old paper records

Decades of property records exist only on paper, in different languages, in different registrar offices. Here is a realistic, phased strategy to digitise them — using existing infrastructure, VLEs, and third-party data companies — without chaos.

⚡ REALISTIC ACKNOWLEDGEMENT

We will not pretend this is easy. Digitising India's urban property records is a **5–10 year effort across 4,000+ urban local bodies**. BPC's Phase 1 does not attempt to digitise everything. It starts with **active transactions and new registrations** — where data quality is cleanest — and works backward into legacy records over time.

01

New Registrations (Immediate)

All new property registrations from Day 1 of pilot get a BPC-ID automatically assigned via Sub-Registrar API. Zero digitisation effort – it's live data.

- > Sub-Registrar API feed
- > RERA new project data
- > Municipal property tax new entries
- > Auto-assigned BPC-ID

02

Recent Records (Year 1-2)

Properties registered in last 10 years – most states already have these in e-Registration systems. BPC pulls via API, cross-checks, assigns BPC-ID.

- > e-Registration DB query
- > RERA registered projects
- > Bank mortgaged properties (NHB data)
- > Validate + assign BPC-ID

03

Legacy Paper Records (Year 2-4)

Pre-digital records – the hardest part. Uses a combination of VLEs, state DILRMP scan data, and third-party digitisation companies under contract.

- > DILRMP scanned records
- > VLE ground-level verification
- > Licensed data companies (Spydra etc)
- > AI OCR + human validation

04

Disputed / Unclear Records (Ongoing)

Records with conflicting data across Registrar, RoR, and municipal systems are flagged – not auto-resolved. Officers review and decide.

- > Conflict flagging engine
- > Officer dashboard for resolution
- > Legal hierarchy (Registrar = final)
- > Court order integration

How VLEs and Third-Party Companies Are Used – and Kept Honest



VLE ROLE

Village Level Entrepreneurs (VLEs) – Ground Data Collectors

VLEs are government-certified operators under CSC (Common Service Centre) scheme. They physically visit properties, scan documents, take GPS-tagged photos, and upload to BPC. **They do not have authority to verify or approve anything** – they are collectors only.

How we prevent VLE fraud: Every upload is tagged with GPS coordinates + timestamp + device ID. Two independent VLE uploads must match before data is accepted. AI cross-checks the scanned document against Registrar records. Any mismatch → human review → VLE flagged.

GPS + Timestamp + Device ID

Dual Upload Verification



THIRD-PARTY DATA COMPANIES

Licensed Data Partners — With a Sunset Clause

Companies like Spydra, Landed, or NoBroker Data hold aggregated property data. BPC may license this for cross-referencing **only** — not as a primary source. Their data is always secondary to Registrar records in our source hierarchy.

Dependency strategy: Phase 1 uses their APIs. Phase 2, BPC mirrors the relevant data internally. Phase 3, BPC's own dataset is primary, third-party becomes optional redundancy. **We never become permanently dependent on a private company for critical data.**

Secondary Source Only

Data Independence Roadmap



SOURCE HIERARCHY (FINAL AUTHORITY)

When records conflict — who wins?

This is the most important rule in the entire system. The source hierarchy is **legally defined and cannot be overridden by any software**:

1. **Court Order** → overrides everything.
2. **Sub-Registrar Record** → final legal title authority.
3. **Revenue Dept / Talathi RoR** → possession and mutation.
4. **RERA Registration** → for new developments.
5. **Municipal Records** → property tax, building permissions.
6. **Third-Party Data** → reference only, never authoritative.
7. **User Upload** → lowest trust, always needs verification.

Hard-coded in System Rules

Cannot be overridden by software

05 — BPC vs DigiLocker: The Comparison

Why BPC is **structurally identical** to DigiLocker — and stronger

The Ministry trusts DigiLocker. Here's an exact side-by-side comparison showing that BPC follows the same architecture — and adds a layer DigiLocker doesn't have.

DigiLocker

National Digital Document Infrastructure —
Trusted, Operational Since 2015

- ✓ Documents stored in govt-controlled cloud
- ✓ Aadhaar-linked login for identity verification
- ✓ IT Act 2000 digital signatures
- ✓ Legally valid — accepted by courts, banks, govt
- ✓ Documents can't be altered once issued
- ✓ Issuer (govt dept) retains source authority
- ✓ Encrypted storage + SSL transmission
- ✗ No cross-department conflict detection
- ✗ No transaction trail linking buyer-seller
- ✗ No fraud flag or risk scoring
- ✗ No audit chain (if dept corrupts source, no check)

BPC — Bharat Property Chain

Property-Specific Digital Infrastructure — Built on
Same Foundations as DigiLocker

- ✓ Documents stored across distributed nodes (BPC + Govt + NIC)
- ✓ Aadhaar OTP + biometric — same as DigiLocker
- ✓ IT Act 2000 digital signatures — same as DigiLocker
- ✓ Legally valid — IT Act + Evidence Act basis
- ✓ Records can't be altered — SHA-256 hash + blockchain log
- ✓ Registrar retains source authority — BPC is read layer
- ✓ AES-256 + TLS 1.3 — stronger than DigiLocker standard
- ✓ **Cross-department conflict detection** — unique to BPC
- ✓ **Transaction trail: token → agreement → registration**
- ✓ **AI fraud flags + risk scoring for banks and buyers**
- ✓ **Blockchain audit chain — even if Registrar data is tampered, log exposes it**

SUMMARY FOR MINISTRY

BPC is **DigiLocker**, but **specifically built for property transactions** — with four additional capabilities that DigiLocker doesn't have because it wasn't designed for them. We are not reinventing a trusted system. We are extending the same trusted architecture into a domain that needs it urgently.

Every hard question. Direct answers.

*These are the questions a Joint Secretary or Minister's technical advisor will ask.
Each has a complete, honest answer.*

Q "Blockchain is not legalized in India. How can you use it?"

Correct — there is no specific "Blockchain Act" in India. However, we are not asking for blockchain to be recognized as a legal title instrument. BPC uses blockchain as a **tamper-evident electronic record system**, which is fully covered under **Section 2(1)(t) of the IT Act, 2000**. Electronic records produced by computers are admissible as evidence under **Section 65B of the Indian Evidence Act, 1872**, provided a certificate is issued. The Supreme Court has upheld this in multiple cases. We are not asking for new law. We are using technology within existing law — exactly as banks, UIDAI, and NPCI do.

Q "What if BPC's servers are hacked? All property records could be stolen or changed."

Three points. **First:** BPC's servers hold a copy of property records — the original, legal, authoritative record always remains with the Sub-Registrar. A hack of BPC's servers does not affect the legal title. **Second:** Our data is AES-256 encrypted at rest and TLS 1.3 encrypted in transmission. Even if the server is physically stolen, the data is unreadable without the decryption keys, which are stored separately in a Hardware Security Module. **Third:** The blockchain audit log is distributed across 3 independent nodes — including a government-controlled node. Changing data on BPC's server without also changing the identical entries on the government's node is impossible — the mismatch is immediately detectable. **This is more secure than the current paper-based system, which can be physically lost, burned, or forged with zero detection.**

Q "Who owns the data? Can BPC sell citizen property data to private companies?"

Citizens own their data. BPC is a data processor, not a data owner — under the **Digital Personal Data Protection Act, 2023 (DPDP Act)**. No property data can be shared with any third party without the explicit digital consent of the property owner. The system has a **Consent Engine** built in — citizens control who sees their data: whether it's a bank, a buyer, or a government officer. BPC signs a **Data Processing Agreement with the Ministry** that legally prohibits commercial sale or misuse of citizen data. Violation is a criminal offence under DPDP.

Act. We welcome a government-appointed data auditor to review our systems at any time.

Q

"Sub-registrar offices still work on paper. How will this practically work?"

BPC does not require Sub-Registrar offices to change their workflow in Phase 1. The integration is **read-only and API-based**. Most states already have e-Registration systems (Maharashtra's iSarita, Karnataka's Kaveri, etc.) — BPC connects to these existing portals via approved APIs to fetch registration data. For offices without APIs, BPC works with the respective state's NIC team to build a lightweight data bridge. We are not asking every Sub-Registrar to open a laptop and use our app. **We work with what exists. We add value on top.**

Q

"Two different states have different land record systems. How does BPC handle this?"

This is the real engineering challenge, and we have a clear answer. BPC uses a **State Adapter Layer** — a set of custom connectors, one per state system. Maharashtra's iSarita produces different data formats than Karnataka's Kaveri or Andhra Pradesh's Meebhoomi. Our adapters normalize all of these into a single BPC data schema. This is how NeSL (National E-Governance Services Limited) operates for IBC insolvency data across multiple states — the same architectural pattern. We pilot in 2 states in Phase 1, build 2 more adapters in Phase 2, and scale state by state. **This is solvable. It just takes time and a state-by-state approach.**

Q

"What happens if BPC as a company shuts down or gets acquired?"

This is a legitimate governance question. Our answer: **the government always holds an independent copy of all BPC audit chain data**. Because the blockchain audit log is distributed and the government's node holds a complete copy, a BPC shutdown does not result in data loss. We also propose a **Data Escrow clause in the Ministry MoU** — requiring BPC to deposit full system backups with NIC every 30 days. Additionally, all the underlying property records are government data to begin with. BPC's unique contribution is the audit chain, transaction trail, and conflict-detection logic — and we commit to open-sourcing this codebase to the government in the event of dissolution.

Minister needs to remember



It Cannot Be Silently Tampered

Every record has a mathematical fingerprint stored in three independent locations — including a government-controlled server. Any change to any record changes the fingerprint and triggers an immediate alert across all three locations. There is no such thing as a "silent" fraud in this system.



It Is Fully Legal Under Current Indian Law

We use no technology that requires new legislation. IT Act 2000, Indian Evidence Act Section 65B, DPDP Act 2023, and RERA 2016 together provide complete legal coverage. The legal record stays with the Sub-Registrar. We add a tamper-proof audit layer on top.



Government Is Always the Final Authority

BPC never auto-approves any property transaction. Technology flags, detects, and assists. But a government-authorized officer holds the final decision. The system is designed to make officers more powerful and informed — not to remove them from the process.

THE ONE COMPARISON THAT CLOSES THE ARGUMENT

India's current paper-based property system can be defrauded with a forged stamp, a corrupt clerk, and ₹50,000. **BPC requires an attacker to simultaneously compromise: three independent servers in different locations, the AES-256 encryption keys held in a Hardware Security Module, the government's own blockchain node, and a government officer's digital signature — all without triggering any of the automated anomaly detection systems.** That is not a realistic attack vector. The question is not "is BPC perfectly secure?" — the question is: "is BPC more secure than paper?" The answer is categorically yes.

Securing India's Property Infrastructure

*Not with promises. With architecture, legal basis, and the same
technology stack that already secures UPI, Aadhaar, and
DigiLocker.*